

Pinching Tactile Display: A Remote Haptic Feedback System for Fabric Texture

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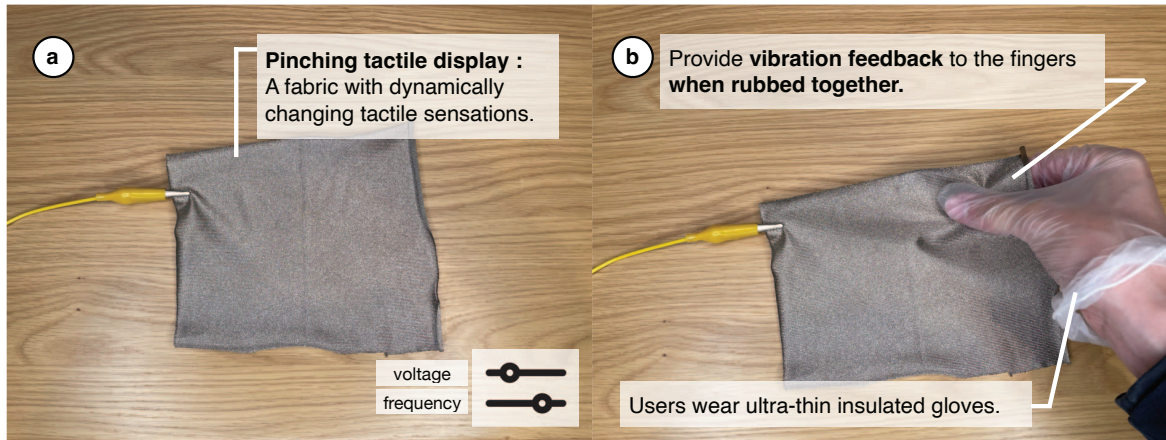


Figure 1: Overview of the Pinch Tactile Display. The fabric changes its tactile sensation by manipulating the electrostatic force between the conductive fabric and the user's finger. Users interact with it by wearing gloves and rubbing them together. The texture variation is achieved by adjusting the voltage and frequency, with the voltage modulated by the power supply and the frequency controlled by the Arduino. The inherent thinness and softness of the fabric allows for a natural touch experience, enhancing immersive interactions in virtual reality environments.

ABSTRACT

In the evolving field of virtual reality (VR) and telepresence, the generation and replication of tactile sensations is critical. Traditional haptic feedback devices have focused on replicating the feel of hard objects, with less attention paid to soft materials. The challenge of reproducing soft tactile sensations stems from two main issues. First, mechanical actuators tend to lose the soft texture when integrated with soft materials. Second, devices such as vibration motors are designed to stimulate a single point, making them unsuitable for reproducing the sensation of more complex tactile interactions, such as the feel of rubbing fabric. We propose Pinching Tactile Display, a system that creates fabric capable of changing its tactile sensation by manipulating the electrostatic force between it and the user's finger. By controlling the voltage and frequency applied to the conductive fabric, different tactile sensations can be dynamically generated. This makes it possible to create a tactile device in which tactile sensations are applied to the entire fabric while maintaining the flexible property of the fabric. As a result, users could experiment with tactile sensations by pinching and rubbing the fabric in the same way they normally touch it. This mechanism has the potential for dynamic tactile transformation of soft materials.

Index Terms: Human-centered computing—Haptic devices; Hardware—Emerging interfaces

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1 INTRODUCTION

In the evolving field of virtual reality (VR) and telepresence, the generation and replication of tactile sensations is critical. Traditional haptic feedback devices have focused on replicating the feel of hard objects [2], with less attention paid to softer materials. Softer materials are used in clothing, bedding and furniture. Their softness and flexibility provide comfort and safety. If the feel of these materials could be tailored to individual preferences, it would make everyday life more comfortable, with potential implications for the fashion industry.

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2 SYSTEM CONFIGURATION

We propose Pinching Tactile Display, a system that produces fabric capable of changing its texture by manipulating the electrostatic force between a conductive fabric and the user's finger. Users wear gloves and rub the cloth together to feel tactile sensations. To change the tactile feel of the cloth, the system adjusts both the voltage and the frequency. This is achieved by altering the supply voltage and by controlling the frequency of the square-wave voltage. This system inherits the natural properties of cloth - thin, soft, and flexible - and provides a tactile sensation in a way similar to touching real fabric, enhancing immersion in virtual reality applications.

2.1 Electro vibration

Figure 2 illustrates the principle of the pinching tactile display. This system utilizes the phenomenon of electrovibration. In the absence

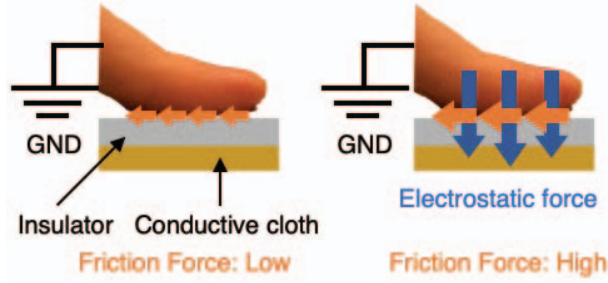


Figure 2: Mechanism of electrovibration.

of high voltage, there is no external force exerted on a finger stroking the tactile display. When voltage is applied to the conductive cloth, the finger connected to GND becomes negatively charged, and the electrodes connected to the high voltage become positively charged. This charge induces electrostatic force on the skin. The periodic variation in frictional force causes periodic deformation of the contacting skin. As a result, users experience a vibrational stimulus.

2.2 Control of Voltage and Frequency

This system alters tactile sensations by varying the magnitude and frequency of the square-wave voltage applied to the conductive cloth. The proposed mechanism varies the voltage applied to the Conductive Cloth by changing the voltage of the DC power source input to the Voltage Booster. The voltage on the Conductive Cloth changes depending on the voltage output by the DC power source and the function by which the Booster of Voltage transforms the input voltage into an output voltage. We used Takasago Seisakusho's LX-2-035-1B compact switching power supply for the stabilized power supply and XP Power's DC-DC converter for the voltage boost.

To control frequency, we used switching action of a transistor to adjust the on/off state of the voltage connected to the conductive cloth, thereby applying a voltage in the form of square waves. The switching of the transistor is controlled by a pulse generator. When the pulse generator applies voltage, the switch turns on; when no voltage is applied from the pulse generator, the switch turns off. When the switch is on, a high voltage increased by the voltage booster is applied to the conductive cloth; when the switch is off, no voltage is applied to the cloth. By equalizing the on and off times, square waves are generated. The frequency of the voltage applied to the Conductive Cloth changes according to the duration of the on and off states of the switch. We used MOSFET TK31N60X by Toshiba for transistor, and Arduino UNO for pulse generator.

2.3 Conductive Fabric and Thin Glove for Insulator

To prevent electricity from flowing from the fabric to the fingers while minimizing the space between the fingers and the fabric, it is essential to allow the user to feel the subtle tactile sensations of the fabric. Distance between the conductive cloth and the finger is a critical factor, so making the thinness of the material a significant factor in achieving changes in tactile sensation. The use of common disposable gloves, which are easily accessible and conform well to the hand, is noteworthy. To ensure that the voltage spreads evenly across all areas, we employed a conductive cloth made of 100% metal fibers with a uniform surface structure. Like typical fabrics, it has the properties of being thin and flexible.

The size of the cloth is approximately $30\text{cm} \times 30\text{cm}$, providing a sufficient area for interaction. It was made entirely of 100% silver fiber, which ensures high conductivity. It had an electrical resistance of $1\ \text{ohm}$. The knitted composition of the fabric allows for stretchability. It weighs $90\text{g}/\text{m}^2$, making it light and easy to handle.

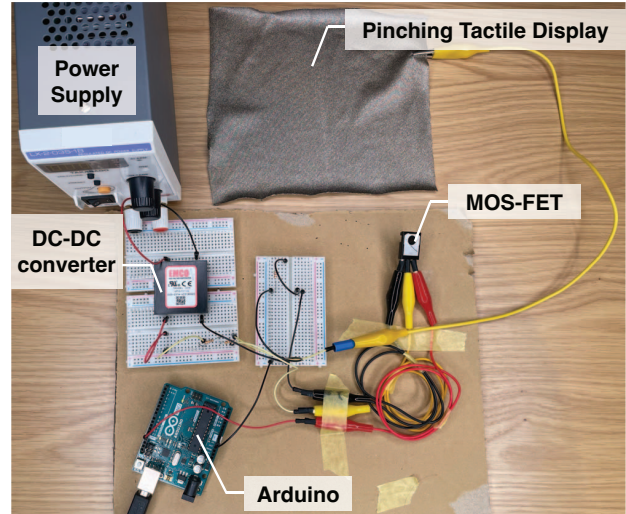


Figure 3: The pinching tactile display configuration, including a conductive cloth, a DC-DC converter, a MOS-FET, An Arduino and a Power Supply.

The thickness of the fabric was 0.232mm . For glove, we used 0.035mm thick, polyvinyl chloride (PVC) gloves that provide effective insulation while maintaining flexibility. The inside of the gloves was powder-free.

2.4 Safety Measures

For safety, the current flowing through the high-voltage part of the tactile device and the circuit connected to it was set very low. The current flowing in the high-voltage section was limited to 0.5mA or less, which is considered safe for the human body [3]. This amount of current is said to be the same as the amount of current that flows through a user's hand when using a capacitive touch screen used in a typical smartphone [1].

3 DISCUSSION

We acknowledge that our current model does not account for the differences in tactile sensations based on the material. In addition, we have not yet evaluated how varying the voltage, frequency, and waveform applied to the conductive fabric can affect tactile sensations. Addressing these issues will be a focus of future research. We aim to use this technology to enhance human interaction with textiles in contexts such as online shopping, wearable technology, and immersive virtual reality experiences.

4 CONCLUSION

In this paper, we proposed Pinching Tactile Display, a system for creating a fabric that can change its tactile sensation by manipulating the electrostatic force between it and the user's finger. By controlling electrostatic force between our finger and the conductive cloth, users can experience multiple tactile sensations of cloth with a single device. We aim to use this system to remotely confirm the tactile sensation of fabrics, particularly in online shopping scenarios, and to enhance interaction within VR environments.

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